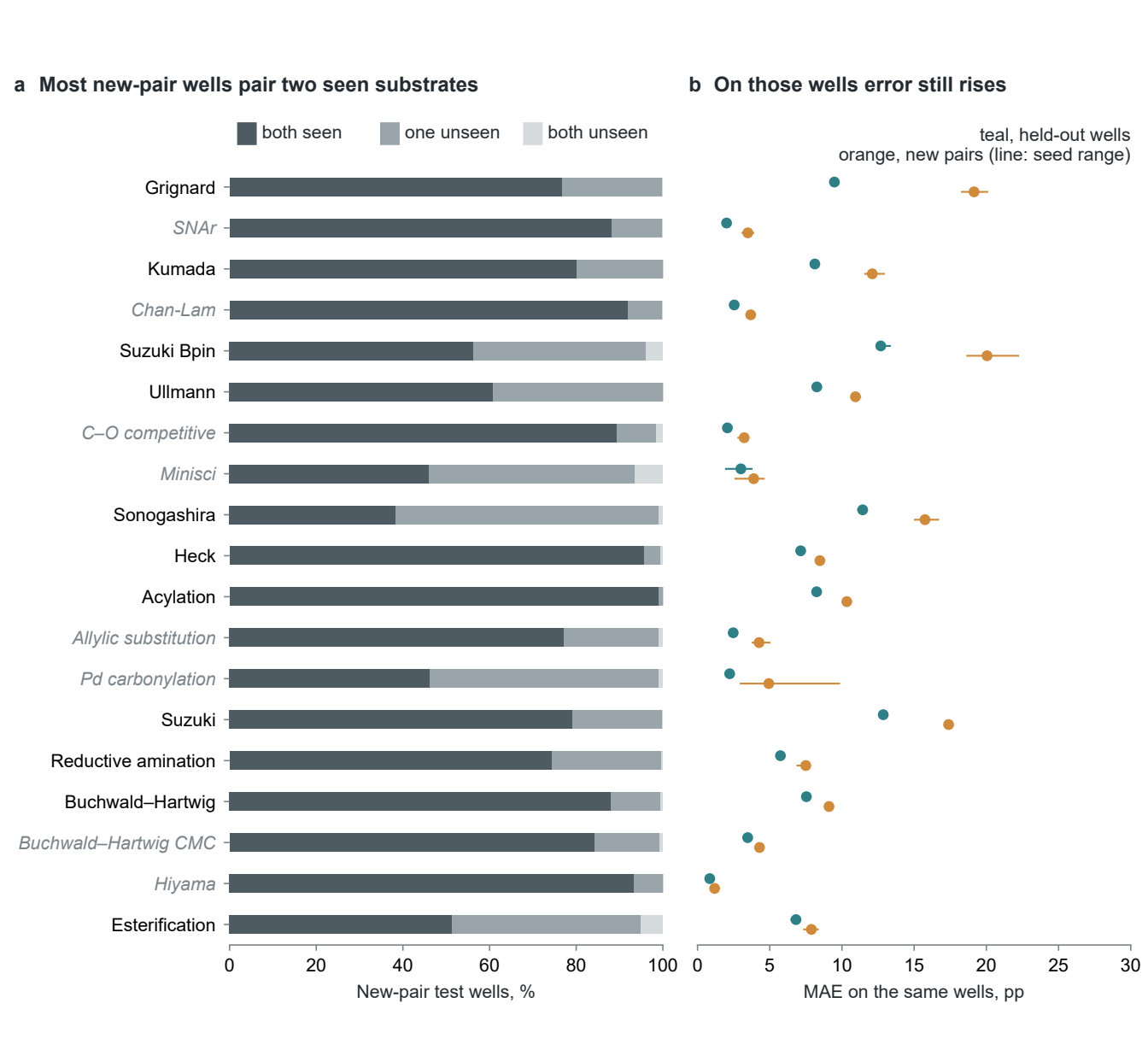

a Most new-pair wells pair two seen substrates
both seen
one unseen
both unseen
Grignard
SNAr
Kumada
Chan-Lam
Suzuki Bpin
Ullmann
C–O competitive
Minisci
Sonogashira
Heck
Acylation
Allylic substitution
Pd carbonylation
Suzuki
Reductive amination
Buchwald–Hartwig
Buchwald–Hartwig CMC
Hiyama
Esterification
0
20
40
60
80
100
New-pair test wells, %
b On those wells error still rises
teal, held-out wells
orange, new pairs (line: seed range)
0
5
10
15
20
25
30
MAE on the same wells, pp